

SECTION AAnswer **all** questions.

1. (a) Write an equation, including state symbols, that represents the enthalpy change of lattice formation of solid magnesium chloride. [1]

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- (b) Complete the following equation: [1]

standard enthalpy
change of solution

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– standard enthalpy
change of lattice formation

2. When chlorine is bubbled into cold sodium hydroxide solution, the following reaction occurs.



- (a) Show that this is a disproportionation reaction. [2]

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- (b) Give **one** common use for sodium chlorate(I). [1]

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3. Calcium hydroxide, $\text{Ca}(\text{OH})_2$, is a strong base that is sparingly soluble in water. A saturated solution of calcium hydroxide has a concentration of $2.34 \times 10^{-2} \text{ mol dm}^{-3}$ at 298 K.

Find the pH of this solution.

[2]

pH =

4. One proposal to reduce the carbon dioxide emissions from public transport is to replace diesel trains with trains powered by hydrogen fuel cells. Give **one other** advantage of using hydrogen fuel cells. [1]

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5. Bismuth is the heaviest naturally occurring element in Group 5. Suggest which oxidation state of bismuth would be most stable. Give a reason for your answer. [2]

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